

Inequality Constraints and Interior-Point Methods

Optimization and Learning for Robot Control

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Review: Equality Constrained Optimization

We previously solved

$$\min_x f(x) \quad \text{s.t.} \quad h(x) = 0$$

Lagrangian:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T h(x)$$

First Order Necessary Conditions:

$$\nabla_x \mathcal{L} = 0$$

$$h(x) = 0$$

This produced a nonlinear system we solved with Newton's method.

Now: Inequality Constraints

General problem:

$$\min_x f(x)$$

$$\text{s.t. } h(x) = 0$$

$$g(x) \leq 0$$

Unlike equality constraints, we cannot directly solve by eliminating constraints.

New conditions appear $\rightarrow$ KKT conditions.

Karush-Kuhn-Tucker (KKT) Conditions

Define Lagrangian:

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$$

KKT Conditions:

1. Stationarity

$$\nabla f(x) + \nabla h(x)^T \lambda + \nabla g(x)^T \mu = 0$$

2. Primal Feasibility

$$h(x) = 0$$

$$g(x) \leq 0$$

3. Dual Feasibility

$$\mu \geq 0$$

4. Complementarity

$$\mu_i g_i(x) = 0$$

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For each constraint:
Either

$$g_i(x) = 0 \quad (\text{active})$$

or

$$\mu_i = 0 \quad (\text{inactive})$$

This makes the problem combinatorial $\rightarrow$ motivates numerical strategies.

1. Active Set Methods

Guess active constraints, solve equality problem, update active set.
Works well when active set small and predictable.

2. Penalty Methods

Replace constraint with penalty:

$$\min_x f(x) + \frac{\rho}{2} \|g(x)_+\|^2$$

Issues: - Ill-conditioning - Large penalty required for accuracy

Improvement: Augmented Lagrangian

Interior-Point (Barrier) Method

Instead of enforcing:

$$g(x) \leq 0$$

Introduce barrier:

$$\min_x f(x) - \mu \sum_i \log(-g_i(x))$$

As $x \rightarrow$ boundary:

$$\log(-g(x)) \rightarrow -\infty$$

Barrier prevents leaving feasible region.

Barrier Optimality Conditions

New objective:

$$\phi(x) = f(x) - \mu \sum_i \log(-g_i(x))$$

Gradient:

$$\nabla \phi(x) = \nabla f(x) - \mu \sum_i \frac{1}{-g_i(x)} \nabla g_i(x)$$

Rewrite:

$$\nabla f(x) + \sum_i \left(\frac{\mu}{g_i(x)} \right) \nabla g_i(x) = 0$$

Define:

$$\mu_i = \frac{\mu}{-g_i(x)}$$

Then:

Primal-Dual Formulation

Instead of pure barrier, introduce dual variable explicitly.

Solve:

$$\nabla f(x) + \nabla g(x)^T \mu = 0$$

$$g(x) \leq 0$$

$$\mu \geq 0$$

$$\mu_i g_i(x) = -\tau$$

As:

$$\tau \rightarrow 0$$

we recover exact KKT conditions.

Slack Variable Reformulation

Convert inequality:

$$g(x) \leq 0$$

Introduce slack:

$$g(x) + s = 0$$

$$s \geq 0$$

Problem becomes:

$$\min_x f(x)$$

$$\text{s.t. } g(x) + s = 0$$

$$s \geq 0$$

Lagrangian with Slack

$$\mathcal{L}(x, s, \mu) = f(x) + \mu^T(g(x) + s)$$

FON conditions:

$$\nabla_x f(x) + \nabla g(x)^T \mu = 0$$

$$g(x) + s = 0$$

$$\mu_i s_i = 0$$

$$\mu_i \geq 0, \quad s_i \geq 0$$

Relaxed Complementarity

Interior point relaxes:

$$\mu_i s_i = \tau$$

Now solve nonlinear system:

$$\nabla f(x) + \nabla g(x)^T \mu = 0$$

$$g(x) + s = 0$$

$$S\mu = \tau e$$

Apply Newton's method to solve this system.

Quadratic Program (QP)

Very important in control:

$$\min_x \frac{1}{2} x^T Q x + q^T x$$

$$\text{s.t. } Ax = b$$

$$Cx \leq d$$

If:

$$Q \succ 0$$

problem is convex $\rightarrow$ unique solution.

QP KKT System

Lagrangian:

$$\mathcal{L} = \frac{1}{2}x^T Qx + q^T x + \lambda^T (Ax - b) + \mu^T (Cx - d)$$

Stationarity:

$$Qx + q + A^T \lambda + C^T \mu = 0$$

Primal:

$$Ax = b$$

$$Cx - d + s = 0$$

Complementarity:

$$\mu_i s_i = 0$$

Why Important for Robot Control?

QP appears in:

- Operational Space Control - Whole-body control - MPC - Inverse dynamics with constraints

Interior-point methods scale well, and are used in solvers like IPOPT.

This is the backbone of modern control.